Productivity & Habit Analytics API (0.1.0)

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A RESTful API for tracking productivity tasks, habits, and personal analytics.

Features

- **Users** — Register and manage user accounts
- **Tasks** — Create and track to-do items with due dates
- **Habits** — Log daily/weekly habits and track streaks
- **Analytics** — Productivity scores, heatmaps, and weekly summaries

Authentication

Most endpoints require a Bearer token. Use `POST /auth/login` to obtain one.

Health

Health

Responses

> **200** Successful Response

GET /health

Response samples

200

Content type
application/json

Copy

```
null
```

Users

Create User

REQUEST BODY SCHEMA: application/json
required

username required	string (Username) [3 .. 30] characters
email required	string <email> (Email)
password required	string (Password) [8 .. 72] characters

Responses

> 201 Successful Response

> 422 Validation Error

POST /users

Request samples

Payload

Content type
application/json

Copy

```
{
    "username": "student_1001",
    "email": "student1001@example.com",
    "password": "Password123"
}
```

Response samples

201 **422**

Content type
application/json

Copy

```
{
    "id": 0,
    "username": "string",
    "email": "user@example.com",
    "created_at": "2019-08-24T14:15:22Z"
}
```

List Users

List users with pagination. Optional search matches username OR email (contains).

QUERY PARAMETERS

`limit`	integer (Limit) Default: `20`

offset	integer (Offset) Default: 0
search >	Search (string) or Search (null) (Search)

Responses

> **200** Successful Response

> **422** Validation Error

GET /users

Response samples

200 422

Content type
application/json

Copy Expand all Collapse all

```
[
 - {
      "id": 0,
      "username": "string",
      "email": "user@example.com",
      "created_at": "2019-08-24T14:15:22Z"
   }
]
```

Get User

PATH PARAMETERS

user_id required	integer (User Id)

Responses

> **200** Successful Response

> **422** Validation Error

GET /users/{user_id}

Response samples

200 422

Content type
application/json

Copy

```
{
    "id": 0,
    "username": "string",
    "email": "user@example.com",
    "created_at": "2019-08-24T14:15:22Z"
}
```

Update User

PATH PARAMETERS

user_id required	integer (User Id)

REQUEST BODY SCHEMA: application/json
required

username >	Username (string) or Username (null) (Username)
email >	Email (string) or Email (null) (Email)

`password` > Password (string) or Password (null) (Password)

Responses

> **200** Successful Response

> **422** Validation Error

PATCH `/users/{user_id}`

Request samples

Payload

Content type
application/json

Copy

```
{
    "username": "student_1001",
    "email": "student1001@example.com",
    "password": "NewPassword123"
}
```

Response samples

200 **422**

Content type
application/json

Copy

```
{
    "id": 0,
    "username": "string",
    "email": "user@example.com",
    "created_at": "2019-08-24T14:15:22Z"
}
```

Delete User

PATH PARAMETERS

`user_id` required	integer (User Id)

Responses

— **204** Successful Response

> **422** Validation Error

DELETE /users/{user_id}

Response samples

422

Content type
application/json

Copy Expand all Collapse all

```
{
  - "detail": [
      + { … }
    ]
}
```

Tasks

Create Task

AUTHORIZATIONS: > *OAuth2PasswordBearer*

REQUEST BODY SCHEMA: application/json
required

title required	string (Title) [1 .. 120] characters
description >	Description (string) or Description (null) (Description)
due_date >	Due Date (string) or Due Date (null) (Due Date)

Responses

> **201** Successful Response

> **422** Validation Error

POST /tasks

Request samples

Payload

Content type
application/json

Copy

```
{
    "title": "Complete Computer Science assignment",
    "description": "Generated from student dataset - Motivation: 8/10",
    "due_date": "2026-03-27"
}
```

Response samples

201 422

Content type
application/json

Copy


```
{
    "id": 0,
    "user_id": 0,
    "title": "string",
    "description": "string",
    "due_date": "2019-08-24",
    "completed": true,
    "completed_at": "2019-08-24T14:15:22Z",
    "created_at": "2019-08-24T14:15:22Z"
}
```


List Tasks

AUTHORIZATIONS: > *OAuth2PasswordBearer*

QUERY PARAMETERS

`completed` >	Completed (boolean) or Completed (null) (Completed)
`due_from` >	Due From (string) or Due From (null) (Due From)
`due_to` >	Due To (string) or Due To (null) (Due To)
`limit`	integer (Limit) Default: `20`
`offset`	integer (Offset) Default: `0`

Responses

> **200** Successful Response

> **422** Validation Error

GET /tasks

Response samples

200 422

Content type
application/json

Copy Expand all Collapse all


```
[
 - {
      "id": 0,
      "user_id": 0,
      "title": "string",
      "description": "string",
      "due_date": "2019-08-24",
      "completed": true,
      "completed_at": "2019-08-24T14:15:22Z",
      "created_at": "2019-08-24T14:15:22Z"
   }
]
```


Get Task

AUTHORIZATIONS: > *OAuth2PasswordBearer*

PATH PARAMETERS

task_id required	integer (Task Id)

Responses

> **200** Successful Response

> **422** Validation Error

GET /tasks/{task_id}

Response samples

200 422

Content type
application/json

Copy

```
{
    "id": 0,
    "user_id": 0,
    "title": "string",
    "description": "string",
    "due_date": "2019-08-24",
    "completed": true,
    "completed_at": "2019-08-24T14:15:22Z",
    "created_at": "2019-08-24T14:15:22Z"
}
```

Update Task

AUTHORIZATIONS: > *OAuth2PasswordBearer*

PATH PARAMETERS

task_id required	integer (Task Id)

REQUEST BODY SCHEMA: application/json
required

title >	Title (string) or Title (null) (Title)

description >	Description (string) or Description (null) (Description)
due_date >	Due Date (string) or Due Date (null) (Due Date)
completed >	Completed (boolean) or Completed (null) (Completed)

Responses

> **200** Successful Response

> **422** Validation Error

PATCH /tasks/{task_id}

Request samples

Payload

Content type
application/json

Copy

```
{
    "title": "Review lecture notes",
    "description": "Focus on chapters 4 and 5",
    "due_date": "2026-03-27",
    "completed": true
}
```

Response samples

200 422

Content type
application/json

Copy

```
{
    "id": 0,
```

```
    "user_id": 0,
    "title": "string",
    "description": "string",
    "due_date": "2019-08-24",
    "completed": true,
    "completed_at": "2019-08-24T14:15:22Z",
    "created_at": "2019-08-24T14:15:22Z"
}
```


Delete Task

AUTHORIZATIONS: > *OAuth2PasswordBearer*

PATH PARAMETERS

`task_id` required	integer (Task Id)

Responses

— **204** Successful Response

> **422** Validation Error

DELETE `/tasks/{task_id}`

Response samples

422

Content type
application/json

Copy Expand all Collapse all

```
{
  - "detail": [
      + { … }
    ]
}
```


Habits

Create Habit

AUTHORIZATIONS: > *OAuth2PasswordBearer*

REQUEST BODY SCHEMA: application/json
required

`name` required	string (Name) [1 .. 100] characters
`frequency`	string (Frequency) `^(daily|weekly)$` Default: `"daily"`

Responses

> **201** Successful Response

> **422** Validation Error

POST `/habits`

Request samples

Payload

Content type
application/json

Copy

```
{
    "name": "Daily Study",
    "frequency": "daily"
}
```

Response samples

201 422

Content type
application/json

Copy

```
{
    "id": 0,
    "user_id": 0,
    "name": "string",
    "frequency": "string",
    "created_at": "2019-08-24T14:15:22Z"
}
```

List Habits

AUTHORIZATIONS: > *OAuth2PasswordBearer*

QUERY PARAMETERS

`limit`	integer (Limit) <= 100 Default: 20
`offset`	integer (Offset) Default: 0

Responses

> **200** Successful Response

> **422** Validation Error

GET /habits

Response samples

200 422

Content type
application/json

Copy Expand all Collapse all


```
[
 - {
      "id": 0,
      "user_id": 0,
      "name": "string",
      "frequency": "string",
      "created_at": "2019-08-24T14:15:22Z"
   }
]
```

Get Habit

AUTHORIZATIONS: > *OAuth2PasswordBearer*

PATH PARAMETERS

`habit_id` required	integer (Habit Id)

Responses

> **200** Successful Response

> **422** Validation Error

GET /habits/{habit_id}

Response samples

200 422

Content type
application/json

Copy

```
{
    "id": 0,
    "user_id": 0,
    "name": "string",
    "frequency": "string",
    "created_at": "2019-08-24T14:15:22Z"
}
```

Update Habit

AUTHORIZATIONS: > *OAuth2PasswordBearer*

PATH PARAMETERS

`habit_id` required	integer (Habit Id)

REQUEST BODY SCHEMA: application/json
required

`name` >	Name (string) or Name (null) (Name)
`frequency` >	Frequency (string) or Frequency (null) (Frequency)

Responses

> **200** Successful Response

> **422** Validation Error

PATCH /habits/{habit_id}

Request samples

Payload

Content type
application/json

Copy

```
{
    "name": "Sleep 8 Hours",
    "frequency": "daily"
}
```

Response samples

200 **422**

Content type
application/json

Copy

```
{
    "id": 0,
    "user_id": 0,
    "name": "string",
    "frequency": "string",
    "created_at": "2019-08-24T14:15:22Z"
}
```

Delete Habit

AUTHORIZATIONS: > *OAuth2PasswordBearer*

PATH PARAMETERS

habit_id required	integer (Habit Id)

Responses

— **204** Successful Response

> **422** Validation Error

DELETE /habits/{habit_id}

Response samples

422

Content type
application/json

Copy Expand all Collapse all

```
{
  - "detail": [
      + { … }
    ]
}
```

Log Habit

AUTHORIZATIONS: > *OAuth2PasswordBearer*

PATH PARAMETERS

habit_id required	integer (Habit Id)

REQUEST BODY SCHEMA: application/json
required

`date` required	string <date> (Date)
`completed`	boolean (Completed) Default: `true`

Responses

> **201** Successful Response

> **422** Validation Error

POST `/habits/{habit_id}/logs`

Request samples

Payload

Content type
application/json

Copy

```
{
    "date": "2026-03-20",
    "completed": true
}
```

Response samples

201 **422**

Content type
application/json

Copy

```
{
    "id": 0,
    "habit_id": 0,
    "date": "2019-08-24",
```

```
    "completed": true
}
```

List Habit Logs

AUTHORIZATIONS: > *OAuth2PasswordBearer*

PATH PARAMETERS

`habit_id` required	integer (Habit Id)

QUERY PARAMETERS

`from_date` >	From Date (string) or From Date (null) (From Date)
`to_date` >	To Date (string) or To Date (null) (To Date)
`limit`	integer (Limit) <= 365 Default: 30
`offset`	integer (Offset) Default: 0

Responses

> **200** Successful Response

> **422** Validation Error

GET /habits/{habit_id}/logs

Response samples

200 422

Content type
application/json

Copy Expand all Collapse all


```
[
 - {
      "id": 0,
      "habit_id": 0,
      "date": "2019-08-24",
      "completed": true
   }
]
```

Delete Habit Log

AUTHORIZATIONS: > *OAuth2PasswordBearer*

PATH PARAMETERS

habit_id required	integer (Habit Id)
log_id required	integer (Log Id)

Responses

— **204** Successful Response

> **422** Validation Error

DELETE /habits/{habit_id}/logs/{log_id}

Response samples

422

Content type
application/json

Copy Expand all Collapse all

```
{
 - "detail": [
     + { … }
   ]
}
```

Auth

Login

REQUEST BODY SCHEMA: application/x-www-form-urlencoded
required

`grant_type` >	Grant Type (string) or Grant Type (null) (Grant Type)
`username` required	string (Username)
`password` required	string <password> (Password)
`scope`	string (Scope) Default: `""`
`client_id` >	Client Id (string) or Client Id (null) (Client Id)
`client_secret` >	Client Secret (string) or Client Secret (null) <password> (Client Secret)

Responses

> **200** Successful Response

> **422** Validation Error

POST `/auth/login`

Response samples

200 422

Content type
application/json

Copy

```
{
    "access_token": "string",
    "token_type": "string"
}
```

Analytics

Analytics Summary

AUTHORIZATIONS: > *OAuth2PasswordBearer*

Responses

> **200** Successful Response

GET `/analytics/summary`

Response samples

200

Content type
application/json

Copy

```
{
    "total_tasks": 0,
    "completed_tasks": 0,
    "completion_rate": 0,
    "total_habits": 0,
    "total_habit_logs": 0
}
```

Analytics Streaks

AUTHORIZATIONS: > *OAuth2PasswordBearer*

Responses

> **200** Successful Response

GET `/analytics/streaks`

Response samples

200

Content type
application/json

Copy Expand all Collapse all

```
[
 - {
      "habit_id": 0,
      "habit_name": "string",
      "current_streak": 0,
      "longest_streak": 0
   }
]
```

Analytics Productivity

AUTHORIZATIONS: > *OAuth2PasswordBearer*

Responses

> **200** Successful Response

GET `/analytics/productivity`

Response samples

200

Content type
application/json

Copy


```
{
   "productivity_score": 0,
   "task_score": 0,
   "habit_score": 0,
   "streak_bonus": 0,
   "total_tasks": 0,
   "completed_tasks": 0,
   "completed_habit_logs": 0,
   "best_current_streak": 0
```

```
}
```

Analytics Weekly

AUTHORIZATIONS: > *OAuth2PasswordBearer*

QUERY PARAMETERS

`start_date` >	Start Date (string) or Start Date (null) (Start Date) Start date (defaults to Monday of current week)
`end_date` >	End Date (string) or End Date (null) (End Date) End date (defaults to Sunday of current week)

Responses

> **200** Successful Response

> **422** Validation Error

GET `/analytics/weekly`

Response samples

200 422

Content type
application/json

Copy Expand all Collapse all

```
{
  "week_start": "2019-08-24",
  "week_end": "2019-08-24",
  "tasks_created": 0,
  "tasks_completed": 0,
  "habit_logs_completed": 0,
  "holiday_count": 0,
```

```
  - "holidays": [
        null
    ]
}
```

Analytics Heatmap

AUTHORIZATIONS: > *OAuth2PasswordBearer*

QUERY PARAMETERS

`days`	integer (Days) `<= 365` Default: `90` Number of past days to include (max 365)

Responses

> **200** Successful Response

> **422** Validation Error

GET `/analytics/heatmap`

Response samples

200 422

Content type
application/json

Copy Expand all Collapse all

```
[
 - {
      "date": "string",
      "activity": 0
   }
]
```